

Getting Ahead of the Flood: Experimental Evidence on Anticipatory Action in Bangladesh and Nepal

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Abstract

As increasingly frequent extreme weather events disrupt lives, institutions are turning to early-warning systems and advance preparation to accelerate aid delivery. Randomized controlled trials in Bangladesh and Nepal tested whether providing cash within days of a flood leads to greater benefits than delivering the same assistance months later—or if it simply shifts the timing of benefits without improving overall welfare. The results suggest that timely cash assistance leads to overall gains in food security and psychosocial well-being. This evidence supports efforts to anticipate crises and release disaster relief quickly.

Keywords: Cash Transfers, Climate Change, Forecasts

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1 Introduction

Anticipatory action is an innovation to the humanitarian aid system. Institutions leverage early-warning weather systems to deliver aid before a predicted hazard occurs or its most severe impacts are felt. When the forecast probability of extreme weather events such as floods exceeds a pre-defined threshold, humanitarian funding is disbursed. This approach represents a significant departure from the more common model of humanitarian relief where support is provided after a disaster strikes. The opportunity to respond quickly to shocks is made possible by dramatic improvements in scientists' ability to predict extreme weather events and better aid distribution systems that reach households more quickly. As of 2024, anticipatory action had become a cornerstone of programming for major institutions including the World Food Programme (WFP), United Nations Office for the Coordination of Humanitarian Affairs (OCHA), and the Food and Agriculture Organization (FAO).

Despite growing enthusiasm for anticipatory action, there is limited causal evidence comparing its effectiveness to humanitarian assistance delivered after disasters occur. This paper examines whether receiving anticipatory action provides additional benefits beyond those that households would have gained, had they received assistance later—or whether it merely shifts those benefits earlier in the season, without generating overall welfare gains. Anticipatory action assistance is disbursed near the peak of a shock and is designed to provide critical support: improving mental health, helping households maintain food security, and allowing them to meet urgent needs without relying on costly coping strategies. After the crisis has passed, when post-flood assistance would typically be disbursed, it is unclear whether anticipatory action continues to provide an advantage or not. Outcomes may be better for households that receive support after the shock because they just received support, while the benefits of anticipatory action may have already faded, suggesting that anticipatory action primarily shifts the timing of support rather than improving overall outcomes. Alternatively, if needs intensify as a crisis unfolds or if market disruptions are more severe in the weeks after a shock, post-flood cash assistance may be more effective. However, if the anticipatory action group experienced short-term benefits that put them on a stronger long-term path, they could experience overall gains from receiving assistance earlier. Finally, as households recover and return to their steady state, further differences between anticipatory action and traditional assistance may emerge. These differences may stem from avoiding the negative consequences of harmful coping strategies or benefiting from time sensitive investments made during the early recovery period.

To examine this question, we experimentally manipulate and study the World Food Programme's (WFP) anticipatory action program. We conducted this study in two countries,

Bangladesh and Nepal, to build a stronger evidence base on the impacts of anticipatory action. In Nepal, we randomly assigned 138 villages to two treatment arms, anticipatory action and a post-flood assistance arm representing the normal timing of transfers delivered by humanitarian assistance. Similarly, in Bangladesh, we randomly assigned 300 villages to the same two treatment groups, (1) anticipatory action and (2) post-flood assistance. In Bangladesh, we extended the design to vary the timing of post-flood assistance.

Both anticipatory action and post-flood assistance groups received a warning about riverine floods, experienced the flood event at the same time, and received a one-time lump sum cash transfer of the same value. The only difference between the two groups was the timing of the cash transfer in relation to the flood event. The anticipatory action group received transfers within days of the flood trigger. In Nepal, approximately 46% of anticipatory action households received transfers 3 days after the flood peak, and over 90% of the anticipatory action group received the transfer within 2 weeks of the flood peak. In Bangladesh, transfers were distributed through mobile money, and the anticipatory action group received their transfers 2 days before floodwaters reached their peak. The post-flood assistance group received transfers much later in both countries—approximately months after the flood peak in Nepal and 1 month after the flood peak in Bangladesh.

We compare outcomes between the anticipatory action group and the post-flood assistance group to assess the impact of anticipatory action relative to post-flood assistance. To measure these effects, we conducted three surveys (referred to as short-run, medium-run, and long-run). Crucially, the short-run survey took place a few weeks after the anticipatory action group received their transfer and after the flood peak, but before the post-flood group received theirs. The medium-run survey was conducted a few weeks after the post-flood group received their transfer, well after the flood peak, such that both groups had received their transfers. Finally, the long-run survey was conducted a few months after the post-flood group received their transfer.

We document four key findings. First, we find that anticipatory action significantly improves food security for households in both countries shortly after the flood, a period of acute stress for households. Pooling data from our short-run surveys reveals that households assigned to anticipatory action have a food consumption score (FCS) that is 1.1 points (2.4%) higher, and are 4.2 percentage points more likely to reach “acceptable” food security levels, driven by more households eating meat. Additionally, the Reduced Coping Strategies Index (rCSI), a proxy of household food insecurity, decreases significantly for the anticipatory action group in both countries by approximately 1.1 points (6.8%). We also find that anticipatory action boosts psychosocial well-being in both countries. We see that our measure of depression (PHQ-4) falls (less depressed) by 0.101 SD, while our measure of life satisfaction

(Cantril ladder) improves (better life) by 0.179 SD. These improvements are larger per dollar of transfer to the effects of multi-faceted anti-poverty interventions, which show an approximate average impact of 0.1 standard deviations per \$1,000 PPP in cash transfers (Ridley et al., 2020). Finally, we observe reduced borrowing and increased savings in Bangladesh (these data were not collected in Nepal), which may help explain the improvements we find in food security and mental health. We do not find evidence of changes in non-agricultural earnings, major non-food expenditures, migration, or agricultural outcomes. This is perhaps unsurprising, as the transfer was not large enough to support major investments, and households likely prioritized meeting immediate needs in the aftermath of flooding.

Second, our results indicate that both groups have similar FCS and rCSI scores in medium run, after the crisis has passed and the post-flood assistance group has received their transfer. This suggests that the anticipatory action group’s receipt of earlier funds improves their food security sooner without a trade-off of (relatively) lower food security in the future. Similarly, we find that the psychosocial well-being of both groups remains comparable at this time. Once again, this suggests that anticipatory action provides early benefits while leading to similar outcomes later.¹

Third, as households recover and return to their steady state, we see no differences between the anticipatory action group and the post-flood assistance group in the longer run. Taken together, these results show that receiving early anticipatory action support leads to overall gains: the total benefits for the anticipatory action group—combining short-run, medium-run, and long-run outcomes—are higher. Households in the anticipatory action group experience positive benefits of the cash transfers just after the floods, but continue to maintain equal or better outcomes than the post-flood group thereafter, suggesting an overall superior trajectory when assistance is delivered earlier than later.

Fourth, we find that the benefits of anticipatory action persist over time. We leverage an extended design in Bangladesh that included a third group left untreated in the medium run, serving as a control at that stage. We find that the anticipatory group continues to experience improvements in food security two months after the flood: their FCS increases by 2.246 points, their RCSI falls by 0.982, and their FIES falls by 0.241, all statistically significant. The post-flood group also shows improvements relative to this temporary control group, but the effects are smaller and less precisely estimated.

This paper makes three primary contributions. First, we contribute to a growing literature on how early assistance helps households cope with extreme weather events. Our study

¹We find no significant changes in the other outcomes we measured, including non-food expenditure, financial outcomes (borrowing, savings, and assets), migration, and non-agricultural earnings. This suggests that early cash assistance is effective for improving food security and mental health but does not drive larger-scale investments or broader economic changes over this time horizon.

is distinctive in its ability to cleanly identify the causal impact of early versus late cash transfers by: (1) explicitly randomizing households into early and late transfer groups; (2) ensuring that the only difference between the two groups is the timing of the transfer, since both groups receive the flood warning when it is issued—allowing us to separate the effects of information from transfer timing; (3) timing our surveys to occur after the late group has received transfers but before the early group has, which is critical for assessing overall welfare gains; and (4) focusing on floods, which are fast-onset, yet increasingly predictable shocks. Crucially, we also measure the impact of anticipatory action in two different country contexts and find remarkably similar results in both, providing strong evidence for the external validity of our findings. Closely related work includes Pople et al. (2021), which studies non-experimental cash transfers but does not include a late-transfer comparison group; Balana et al. (2023), which examines anticipatory action in Nigeria, but does not include a survey round before the late group receives transfers; and Pople et al. (2023), which focuses on droughts rather than floods, which differ in both speed of onset and forecast accuracy.

Second, we contribute to a related literature from predominantly non-emergency settings which has recognized that the timing of programs can affect their impacts (Jalan and Ravallion, 2003; Datt and Ravallion, 1994; Bertrand et al., 2017). Beegle, Galasso, and Goldberg (2015) explicitly shift a cash-for-work program from harvest to lean season, but find no evidence that food security improves. In contrast to these studies, our research explicitly compares pre-shock and post-shock assistance within a humanitarian context.

Finally, we contribute to a global effort to identify programs that help households cope with climate shocks. The literature on the impact of immediate relief efforts in emergency settings is relatively scarce (Jeong and Trako, 2022), and our study offers direct evidence on the effectiveness of cash transfers for humanitarian relief in response to shocks. A broader literature in non-emergency settings—e.g., Emerick et al. (2016), Kondylis et al. (2023), and Jones et al. (2022)—shows that agricultural technologies can improve outcomes in the face of weather shocks, but these interventions are less well-suited for rapid response.

2 Research Design

2.1 Context and Sample

This research takes place in Bangladesh and Nepal, two countries that are particularly vulnerable to natural disasters (Eckstein, Künzel, and Schäfer, 2021). Climate projections suggest that the frequency and intensity of extreme weather events, including heavy monsoon rainfall, will continue to increase in the region (World Bank, 2021). A significant share

of the population in both countries depends on subsistence agriculture (69% in Nepal and 87% in Bangladesh), making their livelihoods highly vulnerable to flooding.

In both countries, WFP recently launched anticipatory action programs. anticipatory action systems use weather forecasts to set thresholds that activate the response to floods. In Nepal, triggers are based on the Global Flood Awareness System (GloFAS) forecast for river floods—which is part of the Copernicus Emergency Management Service of the European Commission (Centre for Humanitarian Data, 2023). Bangladesh uses riverine flood triggers derived from a forecast that includes both GloFAS and national forecasts produced by the Flood Forecasting and Warning Centre of the Bangladesh Water Development Board.² Once triggers are met, institutions (including WFP) can transfer funds directly to households identified as vulnerable using socio-economic measures such as poverty, food insecurity, housing conditions, marginalization, and child malnutrition (stunting and wasting).

2.2 Intervention

The study is conducted in districts that are historically prone to flooding, economically disadvantaged and covered by the GloFAS flood forecasting system. Within these districts, WFP identified a set of villages that had between 25 and 60 registered households eligible to receive transfers in the event of a flood trigger. In Nepal, this process resulted in a sample of 140 villages in the Karnali River Basin. We then surveyed a random sample of 2,212 households evenly distributed across these villages (approximately 15 households per village).³ In Bangladesh, identified 300 villages with large enough numbers of eligible households, from which we surveyed a random sample of 4,721 households (approximately 15 households per village). Tables A.1 and A.2 use WFP administrative data on eligible beneficiaries—collected before transfers were distributed—to show that the majority of households are, as expected, engaged in agriculture (89% in Nepal and 80% in Bangladesh), making them especially exposed to flood shock.

To measure the impact of anticipatory action, we randomly assigned this sample of villages to a group that received anticipatory action, and another that received traditional post-flood assistance. Both groups received warnings about the flood, experienced the flood at the same time, and received transfers of equal value. The only difference between them was the timing of the transfers. Tables A.1 and A.2 confirm that treatment groups are balanced in both countries.

²See Online Appendix B for details on the trigger system.

³Two villages of the original 140 villages selected could not be surveyed because of security concerns, both were assigned the anticipatory action treatment. As a result, our final study sample consists of 68 anticipatory action villages and 70 Post-flood villages.

In Nepal, the flood trigger was met in the Karnali basin on October 8, 2022. Geospatial data from the WFP’s Asset Impact Monitoring System (AIMS) confirmed that extensive flooding occurred in October, affecting approximately 25% of the agricultural land in the survey area. Media reports also indicated that the October 2022 floods were severe, causing flood-related deaths along the river monitored by our study’s flood trigger (BBC, 2022). As a result, households in the anticipatory action group started receiving the one-off transfers a few days after the flood peak in October (delivered at cash points). The post-flood group received transfers approximately months after the flood peak in December. The transfers to both groups were valued at NPR 15,000 (115\$ USD). This represents 80% of the minimum monthly food expenditure basket. Transfers in Nepal were delivered in cash at collection points located nearby targeted villages.

In Bangladesh, the trigger was announced on 4 July 2024. The flooding was significant, and even more severe than in Nepal, with water levels staying above the trigger level for two weeks. Satellite imagery from July 16, 2024, confirms that large parts of the study area still had standing water 12 days after the trigger was announced. Households in the anticipatory action group received the one-off transfers a few days before the flood peak, in late June. The post-flood transfer group received transfers approximately one month after the flood peak, in early August. The transfers to both groups were valued at Tk 5000 (\$41), and they were issued once via mobile money.⁴ This represents 39% of the minimum monthly food expenditure basket. In Bangladesh, some villages were also randomly assigned to a third group, which received transfers with a longer delay—approximately two months after the flood peak and one month after the post-flood group received their transfer. In the short and medium run — before the third group receives any transfers — this setup allows us to assess whether the anticipatory action and post-flood assistance groups experience significant benefits from cash transfers compared to receiving no assistance. Once this third group receives their transfer, we can evaluate whether the anticipatory action and post-flood assistance groups continue to perform as well—or even better—than this last group. In other words, this setup allows us to assess whether the benefits of anticipatory action persist or fade over a longer time horizon than what we can speak to in the Nepal context.

2.3 Considerations when evaluating anticipatory action

Comparing anticipatory action to post-flood assistance The core hypothesis we seek to test is that receiving cash early (anticipatory action) increases welfare relative to

⁴All beneficiaries had an active mobile wallet.

receiving cash after a flood, not the effect of flood exposure.⁵ The expected effect is not clear ex-ante. Indeed, anticipatory action delivers cash within days of a shock, helping households maintain consumption, improve food security, and avoid costly coping strategies—factors that may also reduce stress and support better mental health. While the short-term benefits of receiving assistance immediately following a shock are expected to be positive, it is unclear whether anticipatory action puts households on a stronger long-term trajectory or simply shifts the timing of benefits associated with cash transfers earlier in the season. In the latter case, post-flood groups would eventually experience similar benefits, just at a later point in time. Indeed, receipt of cash transfers may be equally or more beneficial later in the season if conditions worsen in the months following the flood or if financial support is more practical later, when markets have stabilized, and prices are less volatile. The RCT is designed specifically to test whether cash assistance puts households on a better trajectory, holding fixed the set of households who were identified before the trigger.

Anticipatory action and targeting As mentioned, this paper focuses on the benefits of reaching households early by comparing outcomes among for this sample to a comparable sample that was identified before the floods occurred. It is important to note that our experiment does not assess the potential targeting implications of acting earlier versus later. In our study, recipients in both the anticipatory action and post-flood groups are selected before the flood occurs, and a trigger automatically unlocks assistance to all registered recipients. This means we do not capture differences in targeting, and our results reflect the effects of early assistance, net of any targeting variation between approaches.

We also argue that the implications of mis-targeting are likely less consequential for many organizations for two reasons. First, although the benefits of early targeting could be reduced if forecasts are inaccurate and assistance is sent to the wrong households, in our setting the forecasts were accurate—and advances in AI are making these predictions increasingly reliable over time. Second, a potential advantage of later targeting is the ability to verify which households were actually affected by the flood. However, this is often not feasible from an implementation standpoint: implementing partners do not consider it possible to redo vulnerability assessments within the relevant time frame we study, and a one-month delay is insufficient to redefine the sample. Although a longer delay—such as the later post-flood assistance arm in Bangladesh—might, in principle, allow for re-targeting, it introduces additional technical concerns from an evaluation perspective. Re-targeting would require drawing a new study sample that may differ substantially from the original one. For example,

⁵A robust literature uses long panels over large areas to identify the negative impacts of flood exposure (Patel, 2024)

if significant migration has occurred or the village composition has changed in the months following the flood. In that case, the evaluation would end up comparing households that received cash early with a very different set of households receiving cash later, undermining comparability and making it difficult to conduct a credible evaluation. Although we do not observe much attrition in our study sample (Tables A.3-A.6), this does not guarantee that new households did not move into the village who could have been included after re-targeting.

Potential confounds It is also important to note that we identify the impacts of delivering cash assistance in specific months, with flooding occurring at a specific time of the year. This means the treatment effects we identify are “gross” of these seasonal effects, as we do not observe flooding at other times. In Nepal, for example, we compare receiving cash in October—around the time of flooding—to receiving it in December, after the flood. In Bangladesh, we compare receiving cash in June, before the flood, to receiving it in August, after the flood. In both settings, the flood occurred before harvest.⁶ The only way to cleanly separate timing from calendar-month effects would be to run the study over many years in settings where floods occur at different times so that “early” and “late” transfers, and the floods themselves, would fall in many different months. Although this is not feasible in our contexts of seasonal floods, the seasonality of the floods is an integral part of its treatment effects that we measure. That we find very similar effects across two distinct geographies somewhat increases our confidence that the results are not driven by the seasonal placement of our two experiments.

3 Data

3.1 Timeline

Figure 1 overlays the data collection timelines with the time series of water level, as well as the timing of transfers in our two treatment arms. In Nepal, the flood peak occurred on October 9th and the anticipatory action group received transfers starting from October 11th, while the post-flood group received transfers approximately months later in late November. We conducted three rounds of surveys. We first interviewed households in early November

⁶For example, an agriculture-dependent household who experiences a flood that damages crops just before harvest may experience greater impacts from a transfer than a farm household who experiences floods only after harvests are complete or another household whose livelihood is more protected from floods. More broadly, factors that determine whether early or late transfers have bigger overall impacts on food security depend on how much the flood impacts marginal utility of consumption, how much prices and markets for inputs or other goods are affected, and the return on investments immediately after flood and during recovery from actions like replacing damaged assets or repurchasing agricultural inputs.

2022 (what we call the “short-run”), approximately one month after the anticipatory action group received their transfer, but before the post-flood group received theirs. We ran a second interview in mid-January 2023 (what we call the “medium-run”), approximately three months after the post-flood group received their transfer, thus capturing a period after which both groups had received transfers. We conducted the final survey in late May 2023 (what we call the “long-run”), just before the next main planting season.

In Bangladesh, the trigger was activated on June 18th, 2024, and floodwaters continued to rise and peaked on July 6th. The anticipatory action group received transfers starting July 4th, a few days before the flood peak. The post-flood group received transfers approximately one month later in early August. The last group, which serves as a comparison group, received transfers on September 6th. We conducted three rounds of surveys. We first interviewed households in early July 2024 (the “short-run”), approximately two weeks after the anticipatory action group received their transfer, but before the post-flood group received theirs. We ran a second interview in mid-August 2024 (the “medium-run”), approximately two weeks after the post-flood group received their transfer – such that both the anticipatory action group and post-flood group had received transfers. We conducted the final survey in September 2024 (the “long-run”), approximately 2 weeks after the last group received their transfer.⁷

A reasonable worry would be that survey response could be attenuated when households are dealing with floods and that response rates might be endogenous to whether a household has received cash transfers before a given survey round. In the first round, only the anticipatory action group has received transfers. In this round, we drew a target sample of primary survey respondents in each community from the list of households eligible for transfers, and replaced any households who could not be located for surveys. At this stage differential response associated with treatment would mean that the share of actual respondents from the primary household target list is correlated with treatment. In A.3 and A.4 we find that response rates from the primary sample are high and not correlated with treatment. Ninety-three percent of households in Nepal and 80.0% of households in the primary sample in Bangladesh responded to surveys in Round 1, and we can not reject the hypothesis of equal response rates across treatment arms. In subsequent rounds, we can test whether attrition from the round 1 sample is balanced. In Tables A.5 and A.6, we find that resurvey rates high, from 83-95% in every round. The only significant difference is in Bangladesh in Round 2, where attrition is statistically significantly lower in the Post-Flood group compared to the

⁷The long-run survey in Bangladesh occurred two weeks after this very last transfer to ensure that it would be collected at a similar time interval to the transfer as for other rounds. This means that the timing of the long-run survey varied across countries, occurring six months after the post-flood group received its transfer in Nepal, and months after the post-flood group received their transfer in Bangladesh.

anticipatory action and delayed post-flood group, but the difference is only 2-3 percentage points. We conclude that both initial non-response rates and attrition are modest and not meaningfully affected by treatment assignments.

3.2 Main Outcomes

Although the program could affect a wide range of outcomes, the realized impacts ultimately depend on whether households were affected by flooding and how they used the transfer. News reports described the floods as substantial (BBC, 2022), a characterization that we corroborate with household reports of damage to their farming and health. As shown in Table A.7, in Nepal, 68% of households reported crop damage, 59% reported damage to their homes and 59% reported health impacts. In Bangladesh, 88% of households reported health impacts and 43% reported damage to agricultural activities. We also ask households how they spent the transfer (??). The vast majority report spending it on food and water (70% in Bangladesh and 80% in Nepal), while medicine or health services was another common expense (36% in Bangladesh and 47% in Nepal). These data indicate that flooding disrupted the livelihood and health of households and that households used transfers to meet immediate food and health needs. This motivates our focus on food security and mental health outcomes (we also measure medical expenditures, but we do not find any effect on this margin).

Our measure of food security is the Food Consumption Score (FCS) index that was developed by WFP. To construct this index, we ask about frequency of consumption of nine different food groups, customized to the local contexts. These values were weighted according to their nutritional values. Additionally, we include the consumption strategies module from the reduced Coping Strategies Index (rCSI), which measures whether households used specific strategies to manage food consumption over the past four weeks. Although FCS captures consumption across different food groups, rCSI captures the use of negative coping strategies.⁸ Next, we measure psychological well-being in two ways. First, the PHQ-4 patient health questionnaire for anxiety and depression and self-reported life satisfaction through a Cantril ladder.

We also collect a broader set of outcomes outlined in our pre-analysis plan. This includes non-food consumption expenditure over the past month; financial outcomes including savings, borrowing, and assets; non-agricultural earnings including wages, business profits, and livestock; and migration. We do not find any impacts along these dimensions, likely

⁸While these measures are correlated, they do not track one-to-one, for example, a household could have high rCSI if they reduce their intake from three meals per day to two, even if their FCS remains unchanged because they still eat rice and vegetables at both meals.

because the transfer was too small to support larger financial decisions or investments. All corresponding tables are reported in the Appendix (Tables A.13-A.17).

4 Results

4.1 Estimating the impact of anticipatory action relative to post-flood transfers

In each country, we estimate the impact of anticipatory action relative to post-flood transfers on all outcomes through the following regression equation:

$$y_{ijkt} = \beta_t \text{anticipatoryaction}_j * \text{Round}_t + \theta_{tk} + \epsilon_{ijkt} \quad (1)$$

where y_{ijkt} is an outcome for household i in village j and municipality k measured in survey round t . Our coefficient of interest is β_t , the impact in survey round t of being in a village assigned to receive anticipatory action transfers; the omitted category is the post-flood group. We include fixed effects at the district-round level (θ_{tk}), as district is our randomization strata. We cluster standard errors at the level of villages j . Note, for this section, we drop the later post-flood group arm in Bangladesh to keep the design in both countries as similar as possible. We estimate the impacts of anticipatory action and Post-Flood relative to later Post-Flood group in Nepal in the next sub-section.

In Round 1 (the short-run), the post-flood group has not yet received any cash: β_1 therefore identifies the value of receiving cash early relative to not receiving any cash yet. In Rounds 2 (the medium-run) and 3 (the long-run), the post-flood comparison group has received cash: our estimates then capture the difference between receiving funds earlier versus later. One additional note is that Round 3 was conducted at different times after the flood peak and transfer receipt in Nepal and Bangladesh (eight months post-flood peak in Nepal and three months in Bangladesh), which makes direct cross-country comparisons difficult. However, in both cases, the survey was conducted after the floodwaters had fully receded, months after the flood peak when households may be considered to be returning to a non-active flooding equilibrium (see Figure 1).

Our results table first presents the findings for Nepal, followed by Bangladesh. We can also leverage the fact that we ran the same experiment in both countries to estimate the combined effects of anticipatory action across the two contexts. To this end, we take the estimates from each country and weight them by the inverse of their variance. This approach gives us the average impact for the typical country in our sample, with more precise estimates

being given greater weight. Alternatively, we run a specification that pools the data and includes country fixed effects, which yields the average impact for any individual exposed to the treatment in our sample (not reported). Both estimation strategies produce similar results in this case.

Short-Run: Impacts in the immediate aftermath of the floods Table 1 (Columns 1 and 2) presents the impact of anticipatory action on two key measures of food security in Round 1: the Household Food Consumption Score (FCS) and the Reduced Coping Strategy Index (rCSI).⁹ Panel A shows results for Nepal, Panel B for Bangladesh, and Panel C presents an inverse-variance weighted average of the treatment effects from both countries. Pooling these estimates across countries, we find that the anticipatory action group experiences significantly better food security. Their FCS score is 1.1 points (2.4%) higher than the post-flood group, while their rCSI is 1.1 points lower (6.8%), indicating reduced reliance on negative coping strategies. These trends are broadly consistent across both study locations.¹⁰ In Nepal, we observe a 2.9-point (6%) increase in FCS and a 1 point (5.7%) decrease in rCSI, indicating improved food security. In Bangladesh, while the FCS coefficient is positive, it is not statistically significant. However, as we discuss later, we find a significant improvement in another food security measure collected only in Bangladesh —the Food Insecurity Experience Scale (FIES) —which decreases by 0.27 points (4.5%). Bangladesh also experiences a significant decrease in their rCSI of 1.3 points (8%). In both countries, the changes in FCS are driven by an average (significant) 12% increase in the consumption of meats. This suggests that anticipatory action improved access to one of the most expensive and calorie-dense food groups.

To understand the factors behind these changes in the Food Consumption Score, we break down the index into its individual components (Table A.10). Our analysis reveals significant increases in meat consumption for the anticipatory action group (0.26 days per week, 12%). The gains in meat consumption occurred in both countries, increasing by 0.32 days per week (13%) in Nepal and 0.16 days per week (7%) in Bangladesh. This suggests that anticipatory action improved access to one of the most expensive and calorie-dense food groups. One way to benchmark the size of the overall FCS effect is to assess the change in the proportion of people who achieve the “acceptable” standardized threshold of FCS. Rates of acceptable food security in the post-flood group in this round when they had not received transfers were 73.7% in Nepal and 77.54% in Bangladesh. anticipatory action increases the

⁹While round-by-round coefficients for reach outcome are reported across columns, they are estimated using a single regression per equation 1 above.

¹⁰Note that in Bangladesh, the transfer value was less than half the value of transfers distributed in Nepal, so on a per-dollar of transfer basis the food security impacts are similar across countries.

share of households whose FCS exceeded this threshold by 4.2 percentage points across each country, or 8.3 percentage points in Nepal and 2.2 percentage points in Bangladesh.

We now examine how the early distribution of cash affects mental health in the immediate aftermath of a shock, focusing on two measures: PHQ4, a screening tool for anxiety and depression risk, and self-reported life satisfaction using the Cantril Ladder. Table 2, Columns 1 and 2, show the impacts in this round. Panel C presents the pooled estimates where we find that PHQ4 falls (less depressed) by 0.101 standard deviations, while life satisfaction increases (more satisfied) by 0.179 standard deviations. These effect sizes are quite large relative to what others find in the literature. A meta-analysis by (Ridley et al., 2020), show an average impact of anti-poverty programs on PHQ9 of 0.1 standard deviations per \$1,000 PPP. The anticipatory action program achieves the same impact with only a tenth of the transfer size.

We also collected a series of secondary outcomes variables, including Financial behavior (Table A.13) and Non-agricultural earnings (Table A.14). The only statistically significant results in this round are a positive effect on financial behaviors (only collected in Bangladesh in Round 1), where the anticipatory action group saved 3 USD (37%) more and borrowed 9 USD (17%) less than the post-flood group who had not yet received transfers. In a context where thin credit markets lead to high borrowing costs and financial stress, this boost in savings and avoided borrowing costs could be a channel to the improvements we find in food security and mental health. We find no differences on wage income or business income. These comparisons suggest that while cash assistance arriving as early as possible through anticipatory action is effective for improving food security and mental health, and impacts savings and borrowing, it does not lead recipients to change their other economic behaviors or income generation strategies as they cope with floods in the short term.

Medium Run: Relative impacts of early transfers compared to recently delivered post-flood transfers in the early recovery period

This round measures outcomes for both groups shortly after the post-flood group has received transfers. This is now a comparison of the anticipatory action group who received cash as quickly as possible after a flood trigger against a group who has received cash more recently to this survey round. This period is particularly important for evaluating the effectiveness of anticipatory action relative to post-flood transfers. Given that the post-flood group has received cash recently, we might expect that higher food security and better mental health among this group relative to anticipatory action. In this case, the impact of anticipatory action relative to a post-flood transfer would simply be moving the benefits of cash receipt earlier without an absolute gain across all time periods. However, if instead the post-flood group reports similar outcomes

as the anticipatory action households (or remains behind them), then this implies that anticipatory action improves well-being immediately with no relative cost in the future. Therefore, anticipatory action would represent a pure welfare gain relative to post-flood transfers as it puts households on a better trajectory.

Table 1 (Columns 3 and 4) compares food security outcomes between the anticipatory action and post-flood groups in this round, after both groups received their transfers. We find no significant differences between the groups for either measure of food security. The pooled coefficient for FCS is positive but not statistically significant, while it is both positive and borderline-statistically significant ($p = 0.121$) for Bangladesh. While the post-flood group recently received their transfer, they show no noticeable improvement in food security compared to the anticipatory action group, which received their transfer weeks or even months earlier. Equal food security in this round suggests an overall benefit for the anticipatory action group who earlier had higher food security and now achieves equivalence.

We next look at mental health outcomes (Columns 3 and 4 of Table 2). In both the pooled and country-specific estimates, we now cannot reject that mental health and subjective well-being are the same in both groups, again indicating that post-flood assistance merely allows recipients to catch up to, rather than surpass the levels achieved by the anticipatory action group.

Turning to other outcomes, we again find small differences between the anticipatory action group and the post-flood group. Table A.15 shows the impact of anticipatory action on non-food expenditure (newly collected in this round). We find a negative effect in Nepal, where the anticipatory action group spent 15 USD ($p = 0.092$) less on non-food items compared to the ex-post group. This difference is primarily due to 11.2 USD lower expenditures on house repairs. Since we did not collect this data in the previous round, we cannot rule out the possibility that the anticipatory action group in Nepal made house repairs immediately after receiving transfers, while the post-flood group delayed such expenditures.

On financial behaviors (Table A.13) and non-agricultural livelihoods (Table A.14), the only statistically significant effect we find pooled across rounds is a 4 USD (11%) increase in wage income in the anticipatory action group compared to the post-flood group ($p = .085$). In Nepal, we also find a marginally significant decrease in the anticipatory action group on business profits ($p = .099$), but these earnings are small in this round with an estimate of impact of -2.5 USD or 38% of the mean earnings across countries of 6.6 USD. Finally, migration outcomes were only collected in Bangladesh in this round (Table A.17), and we find no effects.

Long-Run: Relative impacts of early transfers compared to post-flood transfers during a later recovery period Our final assessment involves comparisons of anticipatory action against post-flood response in a round at least three months after the flood peak.¹¹

Columns 5 and 6 of Table 1 compare the anticipatory action and post-flood groups in this round on food consumption and coping strategies. We see no effects for FCS in either country, suggesting that the food security gains from anticipatory action have diminished over time. Notably, anticipatory action outcomes remain at least as good as those of the post-flood group, indicating that early assistance did not lead to worse long-term food security outcomes. Indeed, in Bangladesh, where rCSI was still collected in this round, we continue see a statistically significant reduction (4%) in negative coping strategies among the anticipatory action group relative to the post-flood group. Turning to mental health in columns 5 and 6 of Table 2, we still cannot reject the equivalence of PHQ4 and Cantril ladder in the two groups, suggesting that differences in cash transfer timing do not lead to measurable differences in mental health in the long run.

Finally, in this round, we find no statistically significant impacts on our measures of financial behaviors like borrowing, savings, or assets (Table A.13), non-agricultural livelihoods (Table A.14), non-food expenditure (Table A.15), agriculture activities (Table A.16), or migration (Table A.17) with most point estimates close to zero.

Summary Overall, the results suggest that anticipatory action leads to immediate improvements in food security and mental health. In Bangladesh, we also find that households increase their savings and reduce reliance on costly borrowing. These financial adjustments may help explain the improvements we observe in food security and mental health. When the post-flood group also receives an equivalent transfer, this group merely catches up to the level of the anticipatory action group rather than surpassing them, suggesting that the early realization of impacts for the anticipatory action group is a pure gain relative to post-flood humanitarian transfers. This pattern highlights the advantage of anticipatory action in achieving better outcomes early on, with similar outcomes later.

¹¹Recall the long-run survey was conducted six months after the post-flood group received its transfer in Nepal, and months after the post-flood group received their transfer in Bangladesh. In both countries, we do not find significant effects on these measures, so we do not have reason to believe impacts are sensitive to this choice of interval for the final follow-up.

4.2 Adding a comparison to the short- and medium-term estimates in Bangladesh

In Bangladesh, the research design included a third group that received cash transfers two months after the anticipatory action group and one month after the post-flood group. We refer to this group as the later post-flood group. Importantly, this group serves as a “pure control” for the first and second survey rounds in Bangladesh, allowing us to compare the anticipatory action and post-flood groups to individuals who had not yet received any assistance. We estimate the impacts of anticipatory action and Post-Flood relative to Later Post-Flood with the following regression:

$$y_{ijkt} = \beta_t \text{anticipatoryaction}_j * \text{Round}_t + \gamma_t \text{Postflood}_j * \text{Round}_t + \theta_{tk} + \epsilon_{ijkt} \quad (2)$$

where Postflood_j is the 1 month post-flood group, and the excluded category is now the later post-flood group who receives cash two months after the anticipatory action group, representing the longer humanitarian response timeline. The coefficient β_t represents the impact of anticipatory action relative to the later post-flood group who will receive their transfer two months after the flood. The coefficient γ_t represents the impact of the post-flood group (who receives their transfers 1 month after the flood) compared to the later post-flood group who will receive their transfer two months after the flood. This specification allows us to compare the impact of anticipatory action transfers and recent post-flood transfers, each relative to a group that has not yet received any assistance. Crucially, it helps us assess whether the medium-run similarities we documented above between the anticipatory action and post-flood groups reflect genuine improvements in both, or instead indicate that the effects of anticipatory action have faded while post-flood transfers have limited impact.

Earlier, we found that anticipatory action improved food security compared to the 1-month post-flood group in the short-run. Table 3 further shows that the anticipatory action group also experienced better food security outcomes than the 2-month post flood group: FCS improves (insignificantly) by 0.59 points (1.2%), rCSI falls by 8 points (6.4%) and FIES falls by 0.27 points (4.5%). Similarly we see improvements in both PHQ4 ($p = 0.109$) and the Cantril Ladder ($p = 0.055$) compared to the 2-month post flood group (Table 4). We do not interpret differences between the 1 month post-flood and 2 month later post-flood group in the short run, as neither had received their transfer at that point.

In the medium-run, when the 1-month post-flood group has recently received their transfer but the 2-month later post-flood group has not, the anticipatory action group continues to experience better food security outcomes relative to the 2-month post-flood group: FCS improves by 2.25 points, rCSI falls by 0.98 points and FIES falls by 0.24 points. In fact,

the point estimates in this round are nearly the same magnitude as they were in the first round indicating that the benefits of early cash transfers persist with little to no decline, even six weeks after the anticipatory action transfer. We do not find significant differences in psychological outcomes, however. We also find that the 1-month post-flood group (who had just received transfers) shows improvements in FCS, rCSI, and FIES relative to the 2-month post-flood group (who had not yet received transfers), though these differences are not statistically significant.

These patterns suggest that the improvements in food security that we observe among the anticipatory action group in the medium-run reflect the remarkable persistence of anticipatory action impacts: even two months later, the anticipatory action group continues to show outcomes that are better than a group that has yet to receive transfers, and at least as strong (if not stronger) as those of the post-flood group.

Finally, in the long-run, the 2 month post-flood group has now received their transfer. We do not see clear differences in food security measures, or psychosocial outcomes, either for the anticipatory action or 1 month post-flood group, relative to the 2 month post-flood group. This indicates no lasting differences across groups by the end of data collection.

5 Benchmarking Anticipatory Action against other UCTs

When considering anticipatory action as a policy recommendation, a natural question to ask is: how cost effective are anticipatory cash transfers relative to other unconditional cash transfer (UCT) programs? To answer this, we benchmark the effectiveness of anticipatory action cash transfers against those reported in meta-analyses of UCT programs. For example, Crosta et al. (2024) report the impacts of 115 studies on 72 UCT programs. In these studies, the impact of \$ 100 cash transfers is 0.03 standard deviations on food security and 0.04 standard deviations on psychological well-being. In Tables 1 and 2, we report that anticipatory action has a statistically significant effect on these two measures in round 1, relative to the post-flood group. We can directly compare the effect size of Round 1 with the meta-analysis effect after normalizing to the effect per \$100 transferred and standardizing the food security measure to z-scores. In Round 1, the impact per \$100 transferred in the pooled sample is .11 standard deviations of food security and .1624 for mental health. This suggests that the impact of anticipatory action cash transfers in our setting is 3.7 times greater than that reported in the Crosta et al. (2024) meta-analysis for food security and 4.1 times greater for mental health. Since we cannot reject equality of impacts for the anticipatory action group and the Post Flood Group for the other rounds, we could summarize the benefits of Anticipatory Action cash transfers as 3.7-4.1 times larger than typical UCT programs for

the 2-6 weeks following the floods.

This simple benefit ratio could overstate the benefits of anticipatory action compared to other UCTs if there are direct costs of delivering anticipatory action related to registry updating or trigger identification that are born only when anticipatory action is performed but not other UCTs. The WFP country office in Bangladesh provided a breakdown of program costs for anticipatory action and post-flood support. They estimated that anticipatory action was \$5.90 (12.6%) per recipient more expensive to implement per beneficiary than post-flood response, primarily due to differences in the cost of the forecasting system. Ultimately, the average cost will depend on the number of years the intervention runs, since that determines how many flood events we observe and, in turn, how often the trigger is activated.¹² Using a cost ratio of 1.126 suggests that the benefit cost ratio of anticipatory action to UCT on food security is 3.2 ($3.7/1.126$) or 3.6 for mental health.

There are several reasons why comparing this cost-benefit ratio to UCTs is challenging. First, we do not account for the possibility that anticipatory action may generate higher benefits than UCTs because transfers are delivered at times when the marginal value of consumption is higher. If we were to weight impact estimates by marginal utility of consumption, the benefit-cost ratio would likely be higher. Second, our estimates are measured 2 to 6 weeks after the intervention, while the average UCT covered in Crosta et al. (2024) measures impacts at 82 weeks, when effects may already have attenuated. Finally, the number of channels through which different households translate transfers into impacts may change across periods, when our treatment group comparison is also changing. For example, if households diversify the livelihoods they invest in during later rounds, we may be underpowered to detect average treatment effects across these different outcomes because the only comparison we are able to make is between earlier and later cash transfers as opposed to a pure control as in UCT studies.

Nevertheless, we can conclude that anticipatory action cash transfers are at least as effective (and perhaps 3-4 times as impactful) in improving food security and mental health as typical UCT transfers. Although this benefit is concentrated in the few short weeks after the floods, policymakers may particularly value impacts in those weeks, meaning that anticipatory action has meaningfully large benefits compared to small additional costs compared to delivering other UCTs.

¹²However, we do not learn about these risks for a small number of flood realizations over a few study years and across a small number of contexts. We therefore leave the estimation of this cost to future work.

6 Conclusion

In this paper, we assess the impact of the timing of humanitarian cash transfers in response to a flood in Nepal and Bangladesh. We find that the anticipatory action group, who receives the cash within days of a flood, experiences better food security and mental health just after the flood, compared to a post-flood group which has yet to receive their transfer. After the post-flood group receives their transfers, we see no meaningful differences between the two groups. The fact that the anticipatory action group does not perform worse relative to the post-flood group, even when the post-flood group has received cash much more recently, suggests that anticipatory action provides early benefits without compromising any gains from later transfers. The consistency of the results across countries suggests the impact of anticipatory action may not be unique to a particular context. Future research will need to explore a range of additional outcomes, including trade-offs associated with targeting errors, and the intersection between these anticipatory-action programs with other climate-adaptation measures.

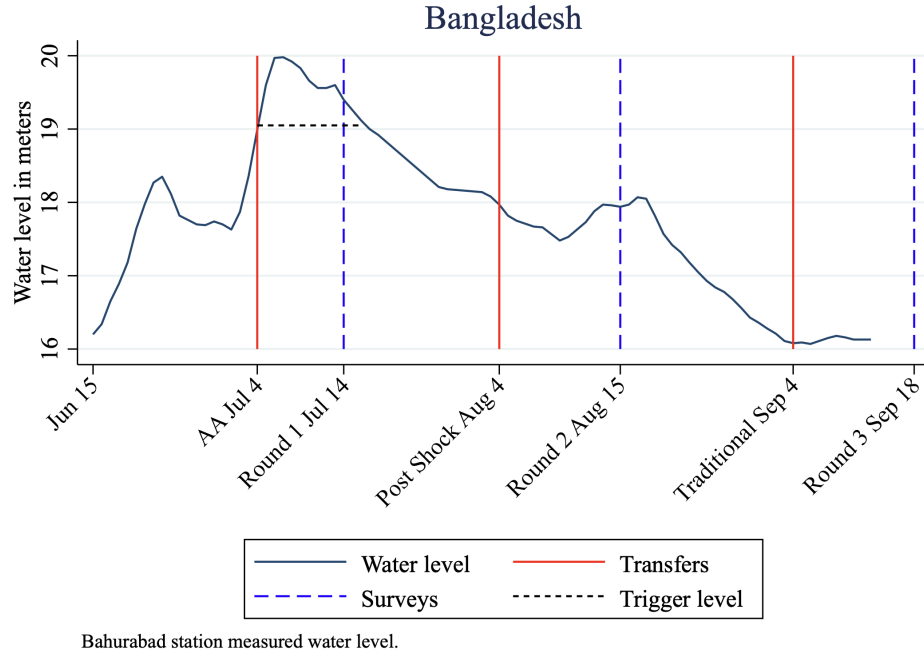
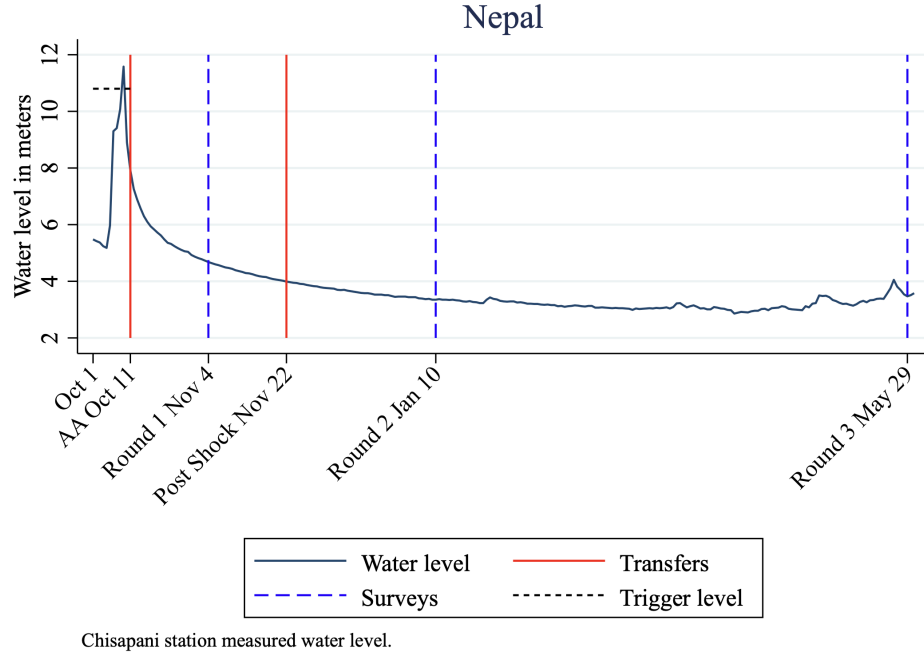
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Figures

Figure 1: Timeline of Flooding and Transfers



Notes: This figure shows a timeline of events in Nepal and Bangladesh respectively. Transfer dates are indicated by solid red lines for the anticipatory action, post-flood group, and status-quo (Bangladesh only) group. Survey dates are indicated by dashed blue lines for all three rounds of surveys. The water level height (in meters) as measured by key monitoring stations is plotted by the solid dark-blue line. The trigger level, which was used to initiate anticipatory action transfer is indicated by the horizontal dashed black line. AA = Anticipatory Action.

Tables

Table 1: Food Security

	Round 1		Round 2		Round 3	
	FCS	rCSI	FCS	rCSI	FCS	rCSI
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Nepal						
Anticipatory Action	2.883**	-0.959*	-0.063	0.220	0.386	
	(1.200)	(0.557)	(1.347)	(0.339)	(1.077)	
	[0.018]	[0.087]	[0.963]	[0.518]	[0.720]	
	N=2212	N=2212	N=2365	N=2365	N=2408	
Panel B: Bangladesh						
Anticipatory Action	0.542	-1.337**	1.168	-0.397	0.455	-0.887**
	(0.707)	(0.562)	(0.751)	(0.566)	(0.547)	(0.429)
	[0.445]	[0.018]	[0.121]	[0.484]	[0.405]	[0.039]
	N=4721	N=4721	N=4761	N=4761	N=4485	N=4485
Panel C: Inverse-Variance Weighted-Average						
Anticipatory Action	1.146*	-1.146***	0.876	0.057	0.441	
	(0.609)	(0.396)	(0.656)	(0.291)	(0.487)	
	[0.060]	[0.004]	[0.182]	[0.846]	[0.365]	
Post-flood mean:NEP	47.9	6.5	47.9	2.6	47.4	
Post-flood mean:BGD	47.0	24.2	48.0	23.4	50.9	21.7
Post-flood mean:Pooled	47.4	16.8	48.0	14.6	49.3	
Strata X Round FE	Yes	Yes	Yes	Yes	Yes	Yes
N Clusters	428	428	438	438	438	438
N Observations	6933	6933	7126	7126	6893	4485

Notes: This table presents the impact of being in the anticipatory action group (β_t) relative to the post-flood group following Equation 1. FCS is Food Consumption Score and rCSI is Reduced Consumption Strategies Index. For each outcome, the coefficients for each round are estimated using a single regression (per Equation 1), but for ease of interpretation we report them in separate columns. An increase in FCS implies greater food security while a reduction in rCSI implies a fall in the number of coping strategies used by the household. Panel A shows results for Nepal only. Panel B shows results for Bangladesh only. Coefficients in Panel C are an invest-variance weighted average of the results from Nepal and Bangladesh shown in Panels A and B, they are not the result of a separate pooled regression. Asterisks indicate statistical significance at the 1% ***, 5% **, and 10% * levels. Post-flood mean is the average value of outcome variable in post-flood group in Nepal (NEP), Bangladesh (BGD) or both countries combined (Pooled).

Table 2: Psychological Wellbeing

	Round 1		Round 2		Round 3	
	PHQ4 (1)	Cantril Ladder (2)	PHQ4 (3)	Cantril Ladder (4)	PHQ4 (5)	Cantril Ladder (6)
Panel A: Nepal						
Anticipatory Action	-0.097 (0.070) [0.167] N=2212	0.154* (0.084) [0.068] N=2212	-0.059 (0.069) [0.395] N=2365	0.084 (0.086) [0.330] N=2365	0.022 (0.061) [0.716] N=2408	0.009 (0.067) [0.890] N=2408
Panel B: Bangladesh						
Anticipatory Action	-0.104* (0.057) [0.068] N=4721	0.195*** (0.066) [0.003] N=4721	0.044 (0.054) [0.411] N=4761	-0.013 (0.043) [0.763] N=4761	-0.014 (0.038) [0.705] N=4485	0.023 (0.035) [0.509] N=4485
Panel C: Inverse-Variance Weighted-Average						
Anticipatory Action	-0.101** (0.044) [0.021]	0.179*** (0.052) [0.001]	0.005 (0.043) [0.901]	0.007 (0.039) [0.865]	-0.004 (0.032) [0.897]	0.020 (0.031) [0.516]
Post-flood mean:NEP	0.051	-0.079	0.030	-0.043	-0.011	-0.004
Post-flood mean:BGD	0.067	-0.103	-0.009	0.022	0.009	0.018
Post-flood mean:Pooled	0.060	-0.093	0.007	-0.006	-0.000	0.009
Strata X Round FE	Yes	Yes	Yes	Yes	Yes	Yes
N Clusters	428	428	438	438	438	438
N Observations	6933	6933	7126	7126	6893	6893

Notes: This table presents the impact of being in the anticipatory action group (β_t) relative to the post-flood group, following Equation 1, on PHQ4 score and Cantril ladder. For each outcome, the coefficients for each round are estimated using a single regression (per Equation 1), but for ease of interpretation we report them in separate columns. PHQ4 and Cantril ladder are standardized to create z-scores using mean and standard deviation within each round. A reduction in PHQ4 implies better mental health while an increase in the Cantril ladder indicates an improvement in life satisfaction. Panel A shows results for Nepal only. Panel B shows results for Bangladesh only. Coefficients in Panel C are an invest-variance weighted average of the results from Nepal and Bangladesh shown in Panels A and B, they are not the result of a separate pooled regression. Asterisks indicate statistical significance at the 1% ***, 5% **, and 10% * levels. Post-flood mean is the average value of outcome variable in post-flood group in Nepal (NEP), Bangladesh (BGD) or both countries combined (Pooled).

Table 3: Food security – Bangladesh only

	Round 1			Round 2			Round 3		
	FCS	rCSI	FIES	FCS	rCSI	FIES	FCS	rCSI	FIES
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Bangladesh									
Anticipatory Action	0.592 (0.806) [0.463]	-1.584** (0.645) [0.015]	-0.274*** (0.089) [0.002]	2.246*** (0.715) [0.002]	-0.982* (0.564) [0.082]	-0.241** (0.097) [0.013]	0.654 (0.547) [0.233]	-0.047 (0.473) [0.920]	-0.032 (0.081) [0.689]
Post-flood Transfer	0.105 (0.842) [0.901]	-0.512 (0.712) [0.472]	-0.120 (0.090) [0.182]	1.078 (0.812) [0.185]	-0.586 (0.551) [0.288]	-0.134 (0.088) [0.128]	0.198 (0.558) [0.723]	0.840* (0.427) [0.050]	0.007 (0.072) [0.924]
P-value AA = post-flood	0.563	0.119	0.101	0.121	0.484	0.283	0.405	0.039	0.622
Delayed post-flood mean	47.123	24.711	6.086	47.479	23.686	5.984	50.809	21.050	5.658
Strata X Round FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N Clusters	300	300	300	300	300	300	300	300	300
N Observations	4721	4721	4721	4761	4761	4761	4485	4485	4485

Notes: This table presents the impact of being in the anticipatory action group (β_t) and the post-flood transfer group (γ_t) relative to the status-quo group following Equation 2. FCS is Food Consumption Score, rCSI is Reduced Consumption Strategies Index, and FIES is Food Insecurity Scale. For each outcome, the coefficients for each round are estimated using a single regression (per Equation 2), but for ease of interpretation we report them in separate columns. An increase in FCS implies greater food security while a reduction in rCSI implies a fall in the number of coping strategies used by the household. Asterisks indicate statistical significance at the 1% ***, 5% **, and 10% * levels. Delayed post-flood mean is the average value of outcome variable in the delayed post-flood group in Bangladesh. AA = Anticipatory Action.

Table 4: Psychological wellbeing – Bangladesh only

	Round 1		Round 2		Round 3	
	PHQ4 (1)	Cantril Ladder (2)	PHQ4 (3)	Cantril Ladder (4)	PHQ4 (5)	Cantril Ladder (6)
Bangladesh						
Anticipatory Action	-0.106 (0.066) [0.109]	0.142* (0.074) [0.055]	-0.041 (0.057) [0.481]	0.009 (0.046) [0.848]	-0.037 (0.037) [0.327]	0.041 (0.035) [0.242]
Post-flood Transfer	-0.004 (0.066) [0.948]	-0.108* (0.064) [0.093]	-0.085 (0.057) [0.135]	0.022 (0.047) [0.641]	-0.022 (0.034) [0.515]	0.018 (0.036) [0.621]
P-value AA = post-flood	0.118	0.001	0.411	0.763	0.705	0.509
Delayed post-flood mean	0.064	-0.023	0.095	-0.008	0.015	-0.026
Strata X Round FE	Yes	Yes	Yes	Yes	Yes	Yes
N Clusters	300	300	300	300	300	300
N Observations	4721	4721	4761	4761	4485	4485

Notes: This table presents the impact of being in the anticipatory action group (β_t) and the post-flood group (γ_t) relative to the status-quo group following Equation 2, on PHQ4 score and Cantril ladder. For each outcome, the coefficients for each round are estimated using a single regression (per Equation 1), but for ease of interpretation we report them in separate columns. PHQ4 and Cantril ladder are standardized to create z-scores using mean and standard deviation within each round. A reduction in PHQ4 implies better mental health while an increase in the Cantril ladder indicates an improvement in life satisfaction. Asterisks indicate statistical significance at the 1% ***, 5% **, and 10% * levels. Delayed post-flood mean is the average value of outcome variable in the delayed post-flood group in Bangladesh. AA = Anticipatory Action.

FAST ACTION FOR FLOODS: RCT EVIDENCE ON FORECAST-BASED CASH TRANSFERS FROM BANGLADESH AND NEPAL

Online appendix

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A Appendix tables and figures

A.1 Balance

Table A.1: Balance – Nepal

Variable	(1) FbAA Mean/(SE)	(2) Post-flood Mean/(SE)	(1)-(2) Difference in means P-value
Has AG job	0.88 (0.02)	0.89 (0.02)	0.74
Has skilled job	0.15 (0.02)	0.21 (0.03)	0.08*
Has pregnant woman	0.04 (0.01)	0.05 (0.01)	0.16
Has disabled person	0.09 (0.02)	0.09 (0.01)	0.68
House brick wall	0.33 (0.04)	0.34 (0.04)	0.74
Receives Welfare	0.27 (0.02)	0.28 (0.01)	0.67
Sampled household	0.87 (0.03)	0.92 (0.01)	0.09*
F-test of joint significance			1.58
Number of observations	1205	1203	2408
Number of clusters	68	70	138

Notes: This table presents descriptive statistics and balance tests across treatment arms using administrative data collected by WFP during registration before floods. Column 1 shows means of variables in the anticipatory action group with standard deviations in parentheses. Column 2 shows means in the post-flood group. Column 3 shows the difference in means. Asterisks in Column 3 indicate statistical significance from a pair-wise t-test on the difference in means at the 1% ***, 5% **, and 10% * levels.

Table A.2: Balance – Bangladesh

Variable	(1)	(2)	(3)	(1)-(2)	(1)-(3)	(2)-(3)
	FbAA Mean/(SE)	Post-flood Mean/(SE)	Delayed Post-flood Mean/(SE)	Difference in means P-value		
Has AG job	0.79 (0.02)	0.80 (0.02)	0.79 (0.01)	0.52	0.83	0.61
Has skilled job	0.02 (0.00)	0.02 (0.00)	0.02 (0.00)	0.60	0.64	0.94
Has pregnant woman	0.15 (0.01)	0.16 (0.01)	0.15 (0.01)	0.38	0.84	0.48
Has disabled person	0.07 (0.01)	0.08 (0.01)	0.07 (0.01)	0.80	0.95	0.86
House brick wall	0.07 (0.01)	0.06 (0.01)	0.07 (0.01)	0.59	0.73	0.83
Evacuated house before	0.29 (0.04)	0.29 (0.03)	0.29 (0.04)	0.93	0.94	0.87
Sampled household	0.75 (0.00)	0.76 (0.01)	0.75 (0.01)	0.47	0.98	0.49
F-test of joint significance				0.27	0.05	0.16
Number of observations	1594	1600	1619	3194	3213	3219
Number of clusters	100	100	100	200	200	200

Notes: This table presents descriptive statistics and balance tests across treatment arms using administrative data collected by WFP during registration before floods in Bangladesh only. Column 1 shows means of variables in the anticipatory action group with standard deviations in parentheses. Column 2 shows means in the Post-flood group. Column 3 shows means in the Delayed Post-flood group. Column 4 shows the difference in means between columns 1 and 2. Column 5 shows the difference in means between columns 1 and 3. Column 6 shows the difference in means between columns 2 and 3. Asterisks in Columns 4-6 indicate statistical significance from a pair-wise t-test on the difference in means at the 1% ***, 5% **, and 10% * levels.

A.2 Attrition

Table A.3: Primary sample response rate by treatment in R1: Bangladesh

	(1) Surveyed R1
FbAA	-0.018 (0.014)
Post-flood	-0.008 (0.013)
Union FE	Yes
Mean in Standard PS group	0.797
Number of observations	4500
R ²	0.023

Notes: This table presents the differences in probability of being surveyed in Round 1 by treatment status in Bangladesh. The sample are the first 15 originally sampled households in 300 villages. The outcome variable takes value of 1 if the sampled household was successfully surveyed by phone and 0 if household was not reachable. Asterisks indicate statistical significance at the 1% ***, 5% **, and 10% * levels.

Table A.4: Primary sample response rate by treatment in R1: Nepal

	(1) Surveyed R1
FbAA	-0.004 (0.030)
Gaupalika FE	Yes
Mean in Standard PS group	0.926
Number of observations	2321
R ²	0.047

Notes: This table presents the differences in probability of being surveyed in Round 1 by treatment status in Bangladesh. The sample are the 2321 originally sampled households in 134 villages. The outcome variable takes value of 1 if the sampled household was successfully surveyed in person and 0 if household was not reachable. Asterisks indicate statistical significance at the 1% ***, 5% **, and 10% * levels.

Table A.5: Attrition after R1: Bangladesh

Variable	(1) AA		(2) Rapid Post		(3) Delayed Post		(1)-(2)		(1)-(3)		(2)-(3)	
	N/Clusters	Mean/(SE)	N/Clusters	Mean/(SE)	N/Clusters	Mean/(SE)	N/Clusters	P-value	N/Clusters	P-value	N/Clusters	P-value
Surveyed R2	1562 100	0.83 (0.01)	1563 100	0.87 (0.01)	1596 100	0.83 (0.01)	3125 200	0.02**	3158 200	0.91	3159 200	0.01**
Surveyed R3	1562 100	0.87 (0.01)	1563 100	0.87 (0.01)	1596 100	0.87 (0.01)	3125 200	0.90	3158 200	0.81	3159 200	0.90
Surveyed R2: Original	1283 100	0.84 (0.01)	1294 100	0.88 (0.01)	1310 100	0.85 (0.01)	2577 200	0.03**	2593 200	0.73	2604 200	0.08*
Surveyed R3: Original	1265 100	0.91 (0.01)	1271 100	0.91 (0.01)	1277 100	0.91 (0.01)	2536 200	0.80	2542 200	0.68	2548 200	0.85

Notes: This table presents the differences in probability of households which were surveyed in Round 1 to be re-surveyed in Rounds 2 and 3 by treatment status in Bangladesh in rows 1-2. The sample are the originally sampled and replacement households in 300 villages which were successfully surveyed in Round 1. The rows 3-4 are restricting to originally sampled 15 households showing up again in Rounds 2 and 3. The outcome variable takes value of 1 if the sampled household was successfully surveyed and 0 if household was not reachable. Asterisks indicate statistical significance at the 1% ***, 5% **, and 10% * levels.

Table A.6: Attrition after R1: Nepal

Variable	(1) Post-flood		(2) FbAA		(1)-(2) Difference in means	
	N/Clusters	Mean/(SE)	N/Clusters	Mean/(SE)	N/Clusters	P-value
Surveyed R2	1138	0.93	1074	0.94	2212	0.46
	66	(0.01)	62	(0.01)	128	
Surveyed R3	1138	0.94	1074	0.95	2212	0.48
	66	(0.01)	62	(0.01)	128	
Surveyed R2: Original	1106	0.93	1034	0.94	2140	0.46
	66	(0.01)	62	(0.01)	128	
Surveyed R3: Original	1106	0.94	1034	0.95	2140	0.54
	66	(0.01)	62	(0.01)	128	

Notes: This table presents the differences in probability of households which were surveyed in Round 1 to be re-surveyed in Rounds 2 and 3 by treatment status in Nepal in rows 1-2. The sample are the originally sampled and replacement households in 128 villages which were successfully surveyed in Round 1. The rows 3-4 are restricting to originally sampled households showing up again in Rounds 2 and 3. The outcome variable takes value of 1 if the sampled household was successfully surveyed and 0 if household was not reachable. Asterisks indicate statistical significance at the 1% ***, 5% **, and 10% * levels.

A.3 Flooding and Use of Funds

Table A.7: Flood damage reported by country

	(1)	(2)
	Nepal	Bangladesh
Flood: health damage	0.49	0.88
Flood: health treatment	0.47	0.62
Flood: house damaged	0.59	0.23
Flood: field crop damage	0.68	0.14
Flood: crop stored damage	0.27	0.01
Flood: livestock damaged	0.39	0.37
Flood: crop cultivation damage	0.69	0.15
Flood: any farm damage	0.79	0.43
Flood: business damage	0.11	0.02
Flood: flood enter house	0.47	0.39
Flood: evacuated house	0.02	0.14
Observations	2365	4485

Notes: This table shows the share of households reporting flood damage in different categories in Nepal (Column 1) and Bangladesh (Column 2).

Table A.8: Cash transfer usage by country

	(1)	(2)
	Nepal	Bangladesh
To buy food or water	0.80	0.70
Medicine or health services	0.47	0.36
To pay back loan	0.24	0.19
Agricultural inputs	0.23	0.09
Festivals and celebrations	0.21	0.00
Education expenses	0.15	0.12
To repair home or asset	0.13	0.03
Rehabilitating crops	0.03	0.01
Saved the money	0.04	0.01
Transport/boat costs	0.01	0.01
Buy clothes	0.00	0.04
Gave to others in need	0.00	0.00
Difficulty withdrawing cash	0.00	0.00
Buy repair/phone	0.00	0.00
Other uses	0.01	0.07
Observations	2359	4485

Notes: This table presents the share of households which used WFP funds for each use case. Data is from Round 2 in Nepal (Column 1) and Round 3 in Bangladesh (Column 2), when all treatment arms have received their transfers. Asterisks indicate statistical significance at the 1% ***, 5% **, and 10% * levels.

A.4 Food Security – additional outcomes

Table A.9: Food Consumption Score (Categorical)

	Round 1			Round 2			Round 3		
	FCS-A (1)	FCS-B (2)	FCS-S (3)	FCS-A (4)	FCS-B (5)	FCS-S (6)	FCS-A (7)	FCS-B (8)	FCS-S (9)
Panel A: Nepal									
Anticipatory Action	0.083*** (0.025) [0.001] N=2212	-0.066*** (0.024) [0.007] N=2212	-0.017*** (0.005) [0.002] N=2212	-0.009 (0.027) [0.734] N=2365	0.007 (0.026) [0.799] N=2365	0.003 (0.003) [0.464] N=2365	0.014 (0.025) [0.581] N=2408	-0.014 (0.024) [0.567] N=2408	0.000 (0.004) [0.994] N=2408
Panel B: Bangladesh									
Anticipatory Action	0.022 (0.017) [0.214] N=4721	-0.015 (0.015) [0.317] N=4721	-0.007 (0.006) [0.267] N=4721	0.012 (0.018) [0.513] N=4761	-0.007 (0.017) [0.686] N=4761	-0.005 (0.003) [0.100] N=4761	0.000 (0.011) [0.985] N=4485	-0.002 (0.011) [0.840] N=4485	0.002 (0.002) [0.265] N=4485
Panel C: Inverse-Variance Weighted-Average									
Anticipatory Action	0.042*** (0.014) [0.004]	-0.030** (0.013) [0.020]	-0.012*** (0.004) [0.002]	0.005 (0.015) [0.726]	-0.003 (0.014) [0.844]	-0.002 (0.002) [0.458]	0.002 (0.010) [0.812]	-0.004 (0.010) [0.674]	0.002 (0.002) [0.317]
Post-flood mean:NEP	0.737	0.240	0.023	0.732	0.262	0.006	0.747	0.246	0.007
Post-flood mean:BGD	0.775	0.198	0.026	0.822	0.169	0.009	0.872	0.125	0.003
Post-flood mean:Pooled	0.759	0.216	0.025	0.784	0.209	0.008	0.816	0.179	0.005
Strata X Round FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N Clusters	428	428	428	438	438	438	438	438	438
N Observations	6933	6933	6933	7126	7126	7126	6893	6893	6893

Notes: This table presents the impact of being in the anticipatory action group (β_t) relative to the post-flood group following Equation 1. FCS-A is an indicator for whether household meets FCS acceptable criteria, FCS-B is an indicator for households falling in the borderline category, and FCS-S is an indicator for whether households are classified as poor. For each outcome, the coefficients for each round are estimated using a single regression (per Equation 1), but for ease of interpretation we report them in separate columns. Panel A shows results for Nepal only. Panel B shows results for Bangladesh only. Coefficients in Panel C are an inverse-variance weighted average of the results from Nepal and Bangladesh shown in Panels A and B, they are not the result of a separate pooled regression. Asterisks indicate statistical significance at the 1% ***, 5% **, and 10% * levels. Post-flood mean is the average value of outcome variable in post-flood group in Nepal (NEP), Bangladesh (BGD) or both countries combined (Pooled).

Table A.10: Food Consumption Score (Disaggregated – Round 1)

	Cereal (1)	Meat (2)	Vegetable (3)	Pulses (4)	Fruit (5)	Milk (6)	Sugar (7)	Oil (8)
Panel A: Nepal								
Anticipatory Action	0.081** (0.037) [0.032] N=2212	0.321*** (0.075) [0.000] N=2212	0.174 (0.169) [0.303] N=2212	0.318 (0.198) [0.111] N=2212	0.015 (0.102) [0.880] N=2212	0.030 (0.156) [0.846] N=2212	0.171 (0.208) [0.412] N=2212	0.172* (0.099) [0.086] N=2212
Panel B: Bangladesh								
Anticipatory Action	0.016 (0.047) [0.741] N=4721	0.160* (0.093) [0.086] N=4721	-0.183 (0.113) [0.107] N=4721	-0.007 (0.065) [0.910] N=4721	0.146 (0.092) [0.113] N=4721	-0.018 (0.057) [0.751] N=4721	-0.012 (0.064) [0.850] N=4721	0.013 (0.051) [0.802] N=4721
Panel C: Inverse-Variance Weighted-Average								
Anticipatory Action	0.056* (0.029) [0.057]	0.258*** (0.058) [0.000]	-0.072 (0.094) [0.444]	0.024 (0.061) [0.697]	0.087 (0.068) [0.201]	-0.012 (0.053) [0.817]	0.004 (0.061) [0.952]	0.046 (0.045) [0.313]
Post-flood mean:NEP	6.820	1.315	4.931	4.218	0.953	1.380	3.597	6.343
Post-flood mean:BGD	6.787	2.857	3.605	2.911	2.409	0.898	0.687	6.646
Post-flood mean:Pooled	6.801	2.207	4.164	3.462	1.795	1.101	1.913	6.518
Strata X Round FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N Clusters	428	428	428	428	428	428	428	428
N Observations	6933	6933	6933	6933	6933	6933	6933	6933

Notes: This table presents the impact of being in the anticipatory action group (β_t) relative to the post-flood group following Equation 1. Each outcome is the number of days in last week where household members ate a food group that features in the FCS score in Round 1. Panel A shows results for Nepal only. Panel B shows results for Bangladesh only. Coefficients in Panel C are an invest-variance weighted average of the results from Nepal and Bangladesh shown in Panels A and B, they are not the result of a separate pooled regression. Asterisks indicate statistical significance at the 1% ***, 5% **, and 10% * levels. Post-flood mean is the average value of outcome variable in post-flood group in Nepal (NEP), Bangladesh (BGD) or both countries combined (Pooled).

Table A.11: Food Consumption Score (Disaggregated – Round 2)

	Cereal (1)	Meat (2)	Vegetable (3)	Pulses (4)	Fruit (5)	Milk (6)	Sugar (7)	Oil (8)
Panel A: Nepal								
Anticipatory Action	-0.028 (0.021) [0.192] N=2365	0.180* (0.103) [0.084] N=2365	0.056 (0.087) [0.522] N=2365	-0.061 (0.219) [0.783] N=2365	0.034 (0.075) [0.653] N=2365	-0.115 (0.181) [0.526] N=2365	-0.311* (0.179) [0.085] N=2365	-0.035 (0.031) [0.259] N=2365
Panel B: Bangladesh								
Anticipatory Action	0.009 (0.029) [0.767] N=4761	0.027 (0.114) [0.813] N=4761	0.024 (0.111) [0.827] N=4761	0.124** (0.059) [0.036] N=4761	0.153 (0.104) [0.141] N=4761	0.104* (0.060) [0.088] N=4761	0.102 (0.078) [0.192] N=4761	0.057 (0.041) [0.165] N=4761
Panel C: Inverse-Variance Weighted-Average								
Anticipatory Action	-0.015 (0.017) [0.377]	0.111 (0.077) [0.147]	0.044 (0.068) [0.521]	0.111** (0.057) [0.050]	0.075 (0.061) [0.219]	0.082 (0.057) [0.154]	0.036 (0.072) [0.615]	-0.002 (0.025) [0.940]
Post-flood mean:NEP	6.983	1.602	6.263	3.121	0.602	1.385	4.641	6.934
Post-flood mean:BGD	6.922	3.340	4.067	2.865	1.344	0.755	0.759	6.764
Post-flood mean:Pooled	6.948	2.602	4.999	2.974	1.029	1.023	2.407	6.836
Strata X Round FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N Clusters	438	438	438	438	438	438	438	438
N Observations	7126	7126	7126	7126	7126	7126	7126	7126

Notes: This table presents the impact of being in the anticipatory action group (β_t) relative to the post-flood group following Equation 1. Each outcome is the number of days in last week where household members ate a food group that features in the FCS score in Round 2. Panel A shows results for Nepal only. Panel B shows results for Bangladesh only. Coefficients in Panel C are an invest-variance weighted average of the results from Nepal and Bangladesh shown in Panels A and B, they are not the result of a separate pooled regression. Asterisks indicate statistical significance at the 1% ***, 5% **, and 10% * levels. Post-flood mean is the average value of outcome variable in post-flood group in Nepal (NEP), Bangladesh (BGD) or both countries combined (Pooled).

Table A.12: Food Consumption Score (Disaggregated – Round 3)

	Cereal (1)	Meat (2)	Vegetable (3)	Pulses (4)	Fruit (5)	Milk (6)	Sugar (7)	Oil (8)
Panel A: Nepal								
Anticipatory Action	-0.001 (0.002) [0.734] N=2408	0.173* (0.088) [0.052] N=2408	-0.022 (0.097) [0.821] N=2408	0.002 (0.161) [0.988] N=2408	0.084 (0.099) [0.397] N=2408	-0.052 (0.153) [0.736] N=2408	-0.364* (0.212) [0.088] N=2408	0.037 (0.030) [0.218] N=2408
Panel B: Bangladesh								
Anticipatory Action	0.004 (0.009) [0.701] N=4485	0.107 (0.076) [0.157] N=4485	-0.129 (0.080) [0.108] N=4485	0.060 (0.072) [0.404] N=4485	0.134** (0.066) [0.043] N=4485	-0.033 (0.066) [0.617] N=4485	-0.088 (0.092) [0.338] N=4485	0.020 (0.022) [0.351] N=4485
Panel C: Inverse-Variance Weighted-Average								
Anticipatory Action	-0.001 (0.002) [0.816]	0.135** (0.057) [0.019]	-0.086 (0.062) [0.165]	0.050 (0.065) [0.441]	0.119** (0.055) [0.030]	-0.036 (0.061) [0.553]	-0.132 (0.084) [0.117]	0.026 (0.017) [0.139]
Post-flood mean:NEP	6.998	1.447	5.797	3.613	1.136	1.259	2.777	6.906
Post-flood mean:BGD	6.976	3.605	4.651	2.870	0.841	1.017	1.777	6.877
Post-flood mean:Pooled	6.986	2.642	5.162	3.202	0.973	1.125	2.223	6.890
Strata X Round FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N Clusters	438	438	438	438	438	438	438	438
N Observations	6893	6893	6893	6893	6893	6893	6893	6893

Notes: This table presents the impact of being in the anticipatory action group (β_t) relative to the post-flood group following Equation 1. Each outcome is the number of days in last week where household members ate a food group that features in the FCS score in Round 3. Panel A shows results for Nepal only. Panel B shows results for Bangladesh only. Coefficients in Panel C are an invest-variance weighted average of the results from Nepal and Bangladesh shown in Panels A and B, they are not the result of a separate pooled regression. Asterisks indicate statistical significance at the 1% ***, 5% **, and 10% * levels. Post-flood mean is the average value of outcome variable in post-flood group in Nepal (NEP), Bangladesh (BGD) or both countries combined (Pooled).

A.5 Pre-specified outcomes outcomes

Table A.13: Borrowing, savings, assets

	Round 1			Round 2			Round 3		
	Borrow (1)	Savings (2)	Assets (3)	Borrow (4)	Savings (5)	Assets (6)	Borrow (7)	Savings (8)	Assets (9)
Panel A: Nepal									
Anticipatory Action				11.099 (30.815) [0.719] N=2365		-0.095 (0.122) [0.434] N=2365	-22.558 (42.031) [0.592] N=2408		-0.101 (0.122) [0.409] N=2408
Panel B: Bangladesh									
Anticipatory Action	-8.786** (4.351) [0.044] N=4140	2.824** (1.109) [0.011] N=4140		7.040 (5.437) [0.196] N=4544	1.423 (2.397) [0.553] N=4544		-5.812 (10.684) [0.587] N=4485	-0.320 (6.257) [0.959] N=4407	-0.020 (0.056) [0.724] N=4485
Panel C: Inverse-Variance Weighted-Average									
Anticipatory Action				7.162 (5.354) [0.181]			-6.829 (10.354) [0.510]		-0.034 (0.051) [0.505]
Post-flood mean:NEP				240.8		-0.1	379.6		0.2
Post-flood mean:BGD	50.4	7.7		67.2	17.6		154.2	48.5	0.0
Post-flood mean:Pooled				142.1			254.8		0.1
Strata X Round FE	Yes	Yes		Yes	Yes	Yes	Yes	Yes	Yes
N Clusters	424	424		438	438	438	438	438	438
N Observations	4140	4140		6909	4544	2365	6893	4407	6893

Notes: This table presents the impact of being in the anticipatory action group (β_t) relative to the post-flood group following Equation 1. Borrow is borrowing since the floods arrived, Savings is total current household savings, and Assets is a count variable of the number of assets owned by the household. Amount borrowed and saved are shown in USD. For each outcome, the coefficients for each round are estimated using a single regression, but for ease of interpretation we report them in separate columns. Outcomes were not collected in all countries in all rounds. Panel A shows results for Nepal only. Panel B shows results for Bangladesh only. Coefficients in Panel C are an invest-variance weighted average of the results from Nepal and Bangladesh shown in Panels A and B (when both are available), they are not the result of a separate pooled regression. Asterisks indicate statistical significance at the 1% ***, 5% **, and 10% * levels. Post-flood mean is the average value of outcome variable in post-flood group in Nepal (NEP), Bangladesh (BGD) or both countries combined (Pooled).

Table A.14: Non-agricultural earnings

	Round 1			Round 2			Round 3		
	Wages (1)	Business (2)	Livestock (3)	Wages (4)	Business (5)	Livestock (6)	Wages (7)	Business (8)	Livestock (9)
Panel A: Nepal									
Anticipatory Action	-2.205 (3.317) [0.507] N=2212	0.590 (1.517) [0.698] N=2212		3.507 (3.676) [0.342] N=2365	-2.487* (1.499) [0.099] N=2365		-7.217 (7.408) [0.332] N=2408	-0.465 (1.124) [0.680] N=2408	-2.526 (7.127) [0.724] N=2408
Panel B: Bangladesh									
Anticipatory Action	-1.849 (2.094) [0.378] N=4721	-0.661 (0.981) [0.501] N=4721		5.244 (3.556) [0.141] N=4761	0.834 (0.963) [0.387] N=4761		-0.377 (2.054) [0.854] N=4485	0.735 (1.111) [0.509] N=4485	3.281 (3.569) [0.359] N=4485
Panel C: Inverse-Variance Weighted-Average									
Anticipatory Action	-1.950 (1.771) [0.271]	-0.292 (0.824) [0.723]		4.404* (2.556) [0.085]	-0.137 (0.810) [0.866]		-0.866 (1.980) [0.662]	0.142 (0.790) [0.858]	2.117 (3.191) [0.507]
Post-flood mean:NEP	44.4	5.3		48.2	6.9		46.5	5.1	58.8
Post-flood mean:BGD	29.1	5.9		31.4	6.3		31.2	10.2	26.0
Post-flood mean:Pooled	35.5	5.7		38.6	6.6		38.0	7.9	40.6
Strata X Round FE	Yes	Yes		Yes	Yes		Yes	Yes	Yes
N Clusters	428	428		438	438		438	438	438
N Observations	6933	6933		7126	7126		6893	6893	6893

Notes: This table presents the impact of being in the anticipatory action group (β_t) relative to the post-flood group following Equation 1. Wage is wage, business, and livestock income earned by the household over the past 30 days. Income is measured in USD. For each outcome, the coefficients for each round are estimated using a single regression, but for ease of interpretation we report them in separate columns. Livestock income was not collected in rounds 1 and 2. Panel A shows results for Nepal only. Panel B shows results for Bangladesh only. Coefficients in Panel C are an invest-variance weighted average of the results from Nepal and Bangladesh shown in Panels A and B (when both are available), they are not the result of a separate pooled regression. Asterisks indicate statistical significance at the 1% ***, 5% **, and 10% * levels. Post-flood mean is the average value of outcome variable in post-flood group in Nepal (NEP), Bangladesh (BGD) or both countries combined (Pooled).

Table A.15: Non-food spending

	Round 1 Spending (1)	Round 2 Spending (2)	Round 3 Spending (3)
Panel A: Nepal			
Anticipatory Action		-14.857* (8.744) [0.092] N=2365	-4.795 (7.813) [0.540] N=2408
Panel B: Bangladesh			
Anticipatory Action		0.754 (2.981) [0.800] N=4761	-2.060 (2.960) [0.487] N=4485
Panel C: Inverse-Variance Weighted-Average			
Anticipatory Action		-0.872 (2.822) [0.757]	-2.403 (2.768) [0.385]
Post-flood mean:NEP		58.9	77.2
Post-flood mean:BGD		46.1	45.2
Post-flood mean:Pooled		51.5	59.5
Strata X Round FE		Yes	Yes
N Clusters		438	438
N Observations		7126	6893

Notes: This table presents the impact of being in the anticipatory action group (β_t) relative to the post-flood group following Equation 1 on non-food consumption. Non-food consumption categories include: electricity, education, fuel, household repairs, medical, and house rent. Values are reported in USD. For each outcome, the coefficients for each round are estimated using a single regression, but for ease of interpretation we report them in separate columns. This outcome was not collected in Round 1 in either country. Panel A shows results for Nepal only. Panel B shows results for Bangladesh only. Coefficients in Panel C are an invest-variance weighted average of the results from Nepal and Bangladesh shown in Panels A and B, they are not the result of a separate pooled regression. Asterisks indicate statistical significance at the 1% ***, 5% **, and 10% * levels. Post-flood mean is the average value of outcome variable in post-flood group in Nepal (NEP), Bangladesh (BGD) or both countries combined (Pooled).

Table A.16: Post-flood agricultural outcomes

Round 3						
	Planted (1)	Area (2)	Costs (3)	Yield (4)	Value (5)	Revenue (6)
Panel A: Nepal						
Anticipatory Action	-0.037 (0.030) [0.209] N=2408	0.007 (0.037) [0.838] N=2408	-2.755 (5.470) [0.615] N=2408	0.001 (0.085) [0.995] N=900	11.625 (22.932) [0.613] N=2408	7.664 (10.409) [0.463] N=2408
Panel B: Bangladesh						
Anticipatory Action	0.005 (0.019) [0.800] N=4485	-0.001 (0.005) [0.892] N=4485	-0.481 (1.393) [0.730] N=4485			
Panel C: Inverse-Variance Weighted-Average						
Anticipatory Action	-0.008 (0.016) [0.637]	-0.001 (0.005) [0.913]	-0.620 (1.350) [0.646]			
Post-flood mean:NEP	0.761	0.330	42.611	1.937	161.684	42.949
Post-flood mean:BGD	0.297	0.048	14.889		36.599	
Post-flood mean:Pooled	0.504	0.174	27.259		92.518	
Strata X Round FE	Yes	Yes	Yes	Yes	Yes	Yes
N Clusters	438	438	438	438	438	438
N Observations	6893	6893	6893	900	6873	2408

Notes: This table presents the impact of being in the anticipatory action group (β_t) relative to the post-flood group on planting crops after the flood affected season following Equation 1. Planted is an indicator for planting any crops. Area is planted in hectares, imputing zeros for those who did not plant. Costs are total expenditure (in USD) on planting activities. Columns 4-6 were only asked in Nepal as households had not finished harvesting post-flood crop in Bangladesh at time of round 3 survey. Yield is output per hectare, Value is the harvest value of crop production in USD, and Revenue is the sale revenue in USD of all crops. Panel A shows results for Nepal only. Panel B shows results for Bangladesh only. Coefficients in Panel C are an invest-variance weighted average of the results from Nepal and Bangladesh shown in Panels A and B, they are not the result of a separate pooled regression. Asterisks indicate statistical significance at the 1% ***, 5% **, and 10% * levels. Post-flood mean is the average value of outcome variable in post-flood group in Nepal (NEP), Bangladesh (BGD) or both countries combined (Pooled).

Table A.17: Migration

	Round 1			Round 2			Round 3		
	Migrated (1)	# (2)	Days (3)	Migrated (4)	# (5)	Days (6)	Migrated (7)	# (8)	Days (9)
Panel A: Nepal									
Anticipatory Action							0.010 (0.025) [0.688] N=2408	0.043 (0.051) [0.409] N=2408	
Panel B: Bangladesh									
Anticipatory Action				-0.005 (0.013) [0.681] N=4761	-0.027 (0.038) [0.482] N=4761	-0.027 (1.539) [0.505] N=576	0.000 (0.012) [1.000] N=4485	0.027 (0.024) [0.266] N=4485	-1.658 (2.301) [0.472] N=565
Panel C: Inverse-Variance Weighted-Average									
Anticipatory Action							0.002 (0.011) [0.864]	0.030 (0.022) [0.173]	
Post-flood mean:NEP							0.553	0.779	
Post-flood mean:BGD				0.127	0.241	18.074	0.126	0.188	29.782
Post-flood mean:Pooled							0.316	0.452	
Strata X Round FE				Yes	Yes	Yes	Yes	Yes	Yes
N Clusters				438	438	438	438	438	438
N Observations				4761	4761	576	6893	6893	565

Notes: This table presents the impact of being in the anticipatory action group (β_t) relative to the rapid post-flood group on migration outcomes following Equation 1. Migrated is an indicator for whether the household has any migrants, # is the number of migrants, and Days is the average duration each migrant was away. In Nepal, respondents were asked about all household members who left the village in the last 12 months and number of days each member spent outside the village. In Bangladesh, respondents were only asked about members who left the house for at least two weeks since June 15 (three weeks before the flood period). Not all outcomes were asked in each round. For each outcome, the coefficients for each round are estimated using a single regression (when multiple rounds were collected), but for ease of interpretation we report them in separate columns. Panel A shows results for Nepal only. Panel B shows results for Bangladesh only. Coefficients in Panel C are an invest-variance weighted average of the results from Nepal and Bangladesh shown in Panels A and B, they are not the result of a separate pooled regression. Asterisks indicate statistical significance at the 1% ***, 5% **, and 10% * levels. Post-flood mean is the average value of outcome variable in post-flood group in Nepal (NEP), Bangladesh (BGD) or both countries combined (Pooled).

A.6 Pre-specified outcomes outcomes - Bangladesh

Table A.18: Borrowing, savings, assets – Bangladesh only

	Round 1			Round 2			Round 3		
	Borrow (1)	Savings (2)	Assets (3)	Borrow (4)	Savings (5)	Assets (6)	Borrow (7)	Savings (8)	Assets (9)
Bangladesh									
Anticipatory Action	-8.568 (5.209) [0.101]	3.227** (1.282) [0.012]		-2.238 (6.208) [0.719]	4.467* (2.382) [0.062]		5.877 (10.430) [0.574]	-2.562 (5.945) [0.667]	0.049 (0.051) [0.341]
Post-flood Transfer	0.443 (5.119) [0.931]	0.820 (1.189) [0.491]		-9.278 (6.050) [0.126]	3.044 (2.049) [0.138]		11.690 (10.750) [0.278]	-2.242 (5.878) [0.703]	0.069 (0.053) [0.199]
P-value AA = post-flood	0.066	0.052		0.196	0.553		0.587	0.959	0.724
Delayed post-flood mean	48.257	6.631		74.817	15.043		137.835	49.539	-0.054
Strata X Round FE	Yes	Yes		Yes	Yes		Yes	Yes	Yes
N Clusters	296	296		300	300		300	300	300
N Observations	4140	4140		4544	4544		4485	4407	4485

Notes: This table presents the impact of being in the anticipatory action group (β_t) and the post-flood transfer group (γ_t) relative to the status-quo group following Equation 2 on financial outcomes. Borrow is borrowing since the floods arrived, Savings is total current household savings, and Assets is a count variable of the number of assets owned by the household. Amount borrowed and saved are shown in USD. For each outcome, the coefficients for each round are estimated using a single regression, but for ease of interpretation we report them in separate columns. Outcomes were not collected in all rounds. Asterisks indicate statistical significance at the 1% ***, 5% **, and 10% * levels. Delayed post-flood mean is the average value of outcome variable in the delayed post-flood group in Bangladesh. AA = Anticipatory Action.

Table A.19: Non-agricultural earnings – Bangladesh only

	Round 1			Round 2			Round 3		
	Wages (1)	Business (2)	Livestock (3)	Wages (4)	Business (5)	Livestock (6)	Wages (7)	Business (8)	Livestock (9)
Bangladesh									
Anticipatory Action	-4.166* (2.395) [0.083]	-0.975 (1.066) [0.361]		2.437 (4.448) [0.584]	0.850 (0.959) [0.376]		-2.715 (2.564) [0.290]	1.163 (1.030) [0.260]	5.369 (3.705) [0.148]
Post-flood Transfer	-4.810* (2.522) [0.057]	-0.650 (1.029) [0.528]		-2.806 (3.167) [0.376]	0.016 (0.970) [0.987]		-2.338 (2.791) [0.403]	0.428 (1.018) [0.675]	2.088 (3.361) [0.535]
P-value AA = post-flood	0.797	0.778		0.141	0.387		0.854	0.509	0.359
Delayed post-flood mean	32.001	7.054		35.462	6.186		32.351	9.804	22.492
Strata X Round FE	Yes	Yes		Yes	Yes		Yes	Yes	Yes
N Clusters	300	300		300	300		300	300	300
N Observations	4721	4721		4761	4761		4485	4485	4485

Notes: This table presents the impact of being in the anticipatory action group (β_t) and the post-flood transfer group (γ_t) relative to the status-quo group following Equation 2 on non-agricultural consumption. Wage is wage, business, and livestock income earned by the household over the past 30 days. Income is measured in USD. For each outcome, the coefficients for each round are estimated using a single regression, but for ease of interpretation we report them in separate columns. Livestock income was not collected in rounds 1 and 2. Asterisks indicate statistical significance at the 1% ***, 5% **, and 10% * levels. Delayed post-flood mean is the average value of outcome variable in the delayed post-flood group in Bangladesh. AA = Anticipatory Action.

Table A.20: Non-food spending – Bangladesh only

	Round 1	Round 2	Round 3
	Spending	Spending	Spending
	(1)	(2)	(3)
Bangladesh			
Anticipatory Action		2.122	1.120
		(3.013)	(2.769)
		[0.482]	[0.686]
Post-flood Transfer		1.368	3.180
		(3.010)	(2.992)
		[0.650]	[0.289]
P-value AA = post-flood		0.800	0.487
Delayed post-flood mean		44.600	41.719
Strata X Round FE		Yes	Yes
N Clusters		300	300
N Observations		4761	4485

Notes: This table presents the impact of being in the anticipatory action group (β_t) and the post-flood transfer group (γ_t) relative to the status-quo group following Equation 2 on non-food consumption. Non-food consumption categories include: electricity, education, fuel, household repairs, medical, and house rent. Values are reported in USD. For each outcome, the coefficients for each round are estimated using a single regression, but for ease of interpretation we report them in separate columns. This outcome was not collected in Round 1. Asterisks indicate statistical significance at the 1% ***, 5% **, and 10% * levels. Delayed post-flood mean is the average value of outcome variable in the delayed post-flood group in Bangladesh. AA = Anticipatory Action.

Table A.21: Post-flood agricultural outcomes – Bangladesh only

	Round 3		
	Planted (1)	Area (2)	Costs (3)
Bangladesh			
Anticipatory Action	0.024 (0.018) [0.186]	0.002 (0.005) [0.693]	1.253 (1.458) [0.391]
Post-flood Transfer	0.019 (0.018) [0.297]	0.003 (0.005) [0.576]	1.734 (1.411) [0.220]
P-value AA = post-flood	0.800	0.892	0.730
Delayed post-flood mean	0.291	0.047	14.182
Strata X Round FE	Yes	Yes	Yes
N Clusters	300	300	300
N Observations	4485	4485	4485

Notes: This table presents the impact of being in the anticipatory action group (β_t) and the post-flood transfer group (γ_t) relative to the status-quo group following Equation 2 on planting crops after flood affected season. Planted is an indicator for planting any crops. Area is planted in hectares, imputing zeros for those who did not plant. Costs are total expenditure (in USD) on planting activities. Asterisks indicate statistical significance at the 1% ***, 5% **, and 10% * levels. Delayed post-flood mean is the average value of outcome variable in the delayed post-flood group in Bangladesh. AA = Anticipatory Action.

Table A.22: Migration – Bangladesh only

	Round 1			Round 2			Round 3		
	Migrated (1)	# (2)	Days (3)	Migrated (4)	# (5)	Days (6)	Migrated (7)	# (8)	Days (9)
Bangladesh									
Anticipatory Action				-0.004 (0.013) [0.759]	-0.041 (0.038) [0.290]	2.637* (1.595) [0.099]	-0.010 (0.012) [0.369]	-0.020 (0.028) [0.471]	-0.231 (2.236) [0.918]
Post-flood Transfer				0.001 (0.014) [0.930]	-0.014 (0.044) [0.753]	1.609 (1.607) [0.318]	-0.010 (0.012) [0.393]	-0.047* (0.028) [0.097]	1.426 (2.516) [0.571]
P-value AA = post-flood				0.681	0.482	0.505	1.000	0.266	0.472
Delayed post-flood mean				0.118	0.254	18.652	0.132	0.219	26.576
Strata X Round FE				Yes	Yes	Yes	Yes	Yes	Yes
N Clusters				300	300	300	300	300	300
N Observations				4761	4761	576	4485	4485	565

Notes: This table presents the impact of being in the anticipatory action group (β_t) and the post-flood transfer group (γ_t) relative to the status-quo group following Equation 2 on migration outcomes. Migrated is an indicator for whether the household has any migrants, # is the number of migrants, and Days is the average duration each migrant was away. In Bangladesh, respondents were only asked about members who left the house for at least two weeks since June 15 (three weeks before the flood period). Not all outcomes were asked in each round. For each outcome, the coefficients for each round are estimated using a single regression (when multiple rounds were collected), but for ease of interpretation we report them in separate columns. Asterisks indicate statistical significance at the 1% ***, 5% **, and 10% * levels. Delayed post-flood mean is the average value of outcome variable in the delayed post-flood group in Bangladesh. AA = Anticipatory Action.

B Flood Forecast and Transfer Details

In Nepal and Bangladesh, responding institutions rely on two triggers: the ‘readiness’ and ‘action’ triggers (Centre for Humanitarian Data, 2023). These triggers are defined based on the Global Flood Awareness System (GloFAS) forecast – which is part of the European Commission’s Copernicus Emergency Management Service and serves as an operational system for predicting and monitoring floods worldwide – in collaboration with national meteorology departments.¹³

In Nepal, the readiness trigger is activated when the 7-day Global Flood Awareness System (GloFAS) forecast predicts a 70 percent likelihood of river discharge surpassing a 1-in-2 year return period level (which is a discharge level expected to occur once every 2 years) (OCHA, 2024). In Bangladesh, the readiness trigger is reached when the water discharge at the Bahadurabad gauging station (located on the Jamuna river in the north of Bangladesh) over a period of three consecutive days is forecasted by the GloFAS model with a maximum 15-day lead time to be more than 50% likely to cross the 1-in-5-year return period (OCHA, 2023). Once this readiness threshold is met, the UN’s Central Emergency Response Fund (CERF) disburses approval letters and initial rounds of funding to responding agencies (including WFP), which they can use to disseminate early warning messages, dispatch initial supplies, and coordinate with local authorities.

Once a second ‘action’ trigger is met, institutions (including the WFP) can transfer funds directly to vulnerable households in the identified areas. In Nepal, this action trigger is met if two conditions are fulfilled: first, an alert from Nepal’s Department of Hydrology and Meteorology (DHM); and second, either the GloFAS 3-day forecast indicates a 70 percent likelihood of river discharge exceeding the 1-in-2-year return period level, or the river’s height exceeds the Nepali government’s pre-defined trigger level (OCHA, 2024). This means that transfers are provided for floods that are worse than the average year.¹⁴ In Bangladesh, the action trigger is fulfilled when the water level at Bahadurabad is forecasted by the government’s Flood Forecast and Warning Center (FFWC) to cross the government-defined “Trigger Level” + 0.85 meters, and GloFAS shows an sustained or increasing trend of water discharge for at least three consecutive days from when the trigger level was crossed (OCHA, 2023). In Nepal (2022), anticipatory action households received transfers a few days after the flood peak, and in Bangladesh (2024), they received them a few days before the peak.

¹³GloFAS employs automated processing of Copernicus Sentinel-1 Synthetic Aperture Radar (SAR) satellite data, utilizing three state-of-the-art satellite flood detection algorithms in an ensemble approach for flood monitoring (GloFAS, 2023).

¹⁴We acknowledge that stricter trigger criteria that activate cash transfers for more severe but less frequent events could yield different outcomes.